Teen Mental Health: Objectives & Guide

Guiding questions:
- How do social media hours and sleep relate to academic performance and well-being indicators?
- Which differences across gender and platform usage appear most notable in this sample?
- What patterns seem consistent versus likely noise at this sample size?

Reading guide:
- Start with the quick KPIs and correlation heatmap for orientation.
- Scan the distribution and group comparisons to understand typical ranges and differences.
- Use the interactive section for ad-hoc deep dives; finish with the actionable summary.

```
Rows x Columns: (1200, 13)
Columns: ['age', 'gender', 'daily_social_media_hours', 'platform_usage', 'sleep_hou
```

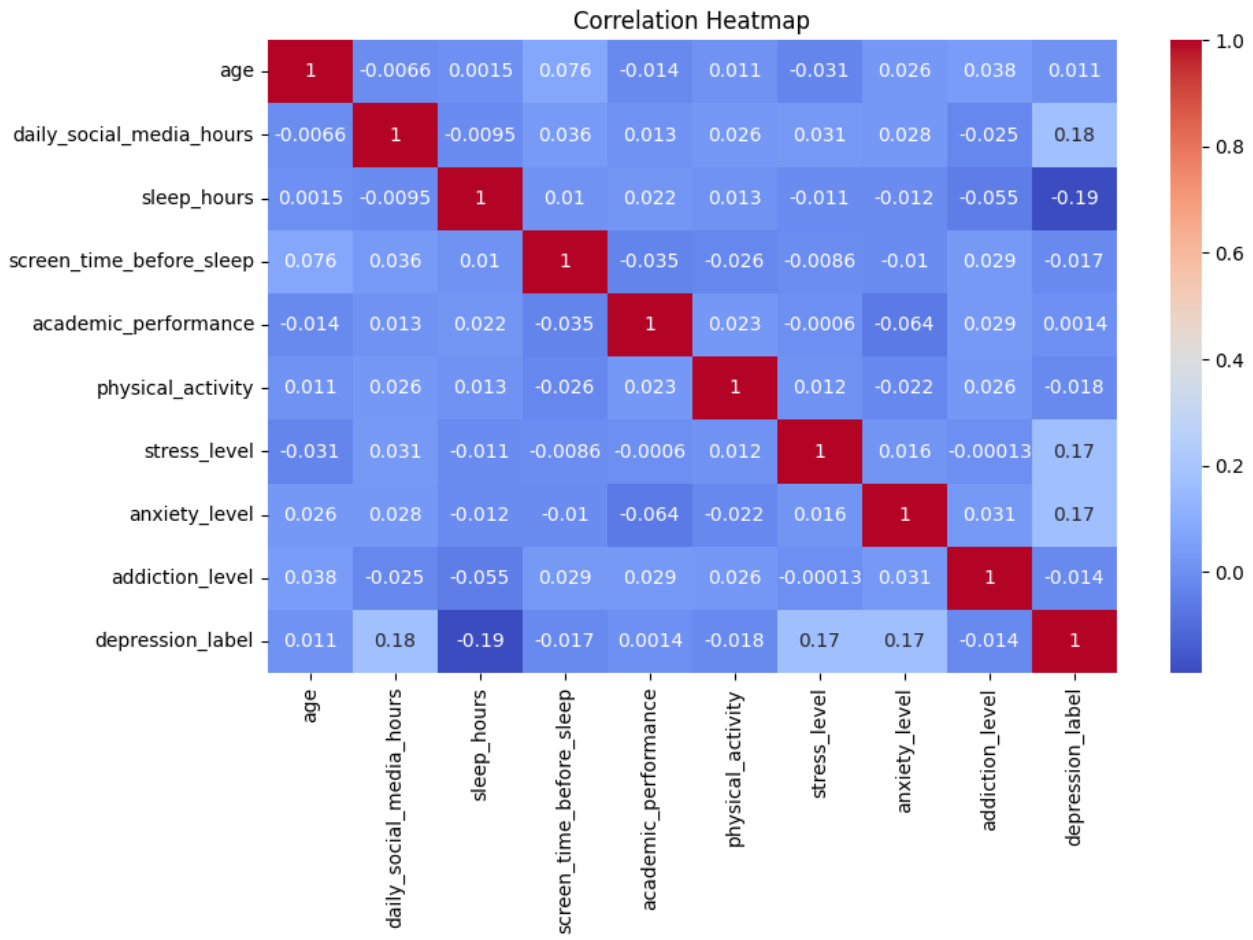
Data context & quality

This dataset captures teens' demographics, habits, and well-being indicators. Basic checks indicate no missing values across columns and numeric ranges that appear sensible for this context (e.g., ages within teens, hours within common bounds). Subsequent summaries and plots are based on the raw dataset without imputation.

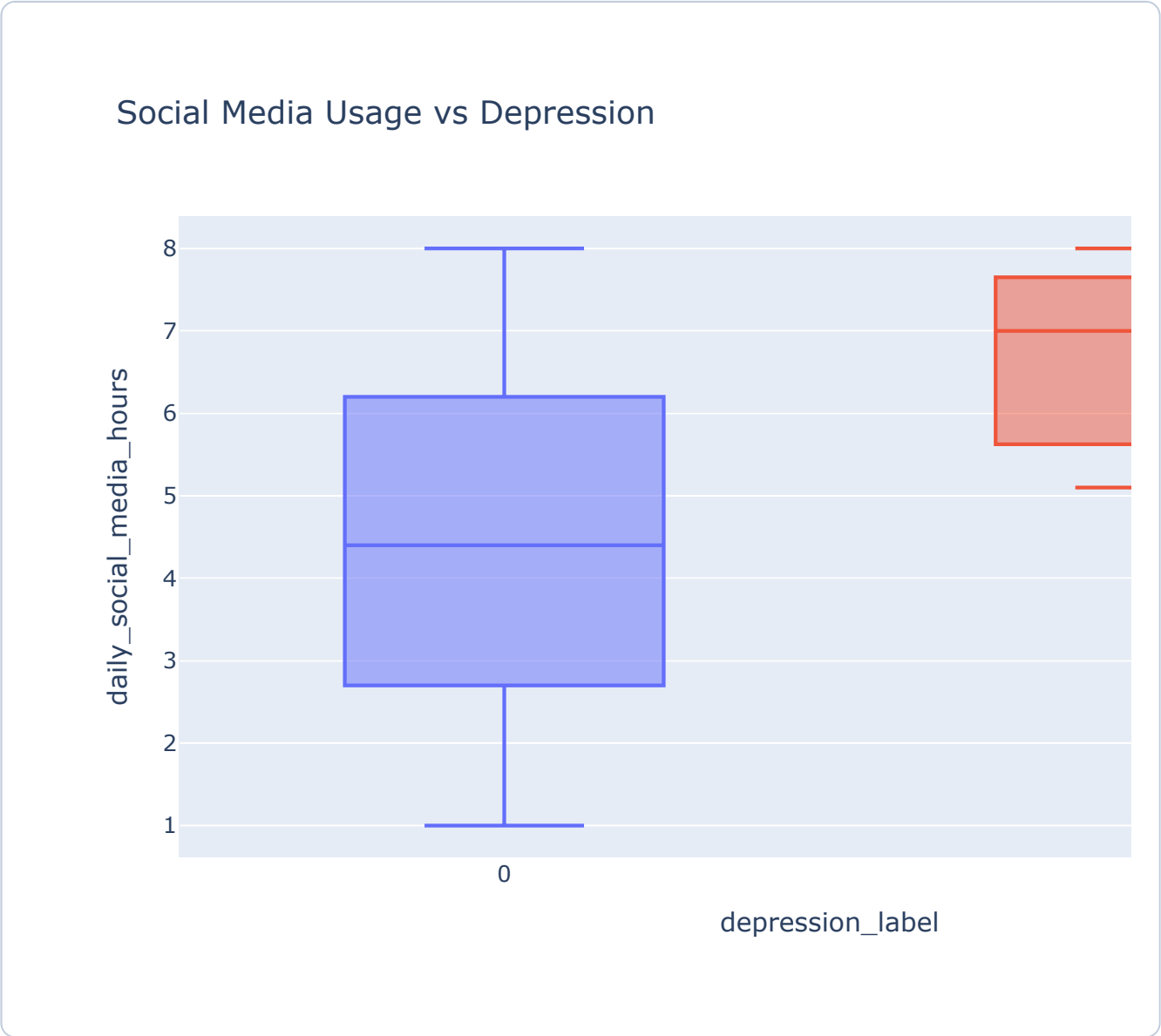
```
mean                    1.740333              2.990383           1.014500
std                     0.716660              0.576758           0.582185
min                     0.500000              2.000000           0.000000
25%                     1.100000              2.500000           0.500000
50%                     1.800000              2.990000           1.000000
75%                     2.400000              3.480000           1.500000
max                     3.000000              4.000000           2.000000

       stress_level  anxiety_level  addiction_level  depression_label
count   1200.000000    1200.000000      1200.000000       1200.000000
mean       5.445833       5.636667         5.565000          0.025833
std        2.903290       2.859453         2.830627          0.158704
min        1.000000       1.000000         1.000000          0.000000
25%        3.000000       3.000000         3.000000          0.000000
50%        5.000000       6.000000         6.000000          0.000000
75%        8.000000       8.000000         8.000000          0.000000
max       10.000000      10.000000        10.000000          1.000000
age                          0
gender                       0
daily_social_media_hours     0
platform_usage               0
sleep_hours                  0
screen_time_before_sleep     0
academic_performance         0
physical_activity            0
social_interaction_level     0
stress_level                 0
anxiety_level                0
addiction_level              0
depression_label             0
dtype: int64
```

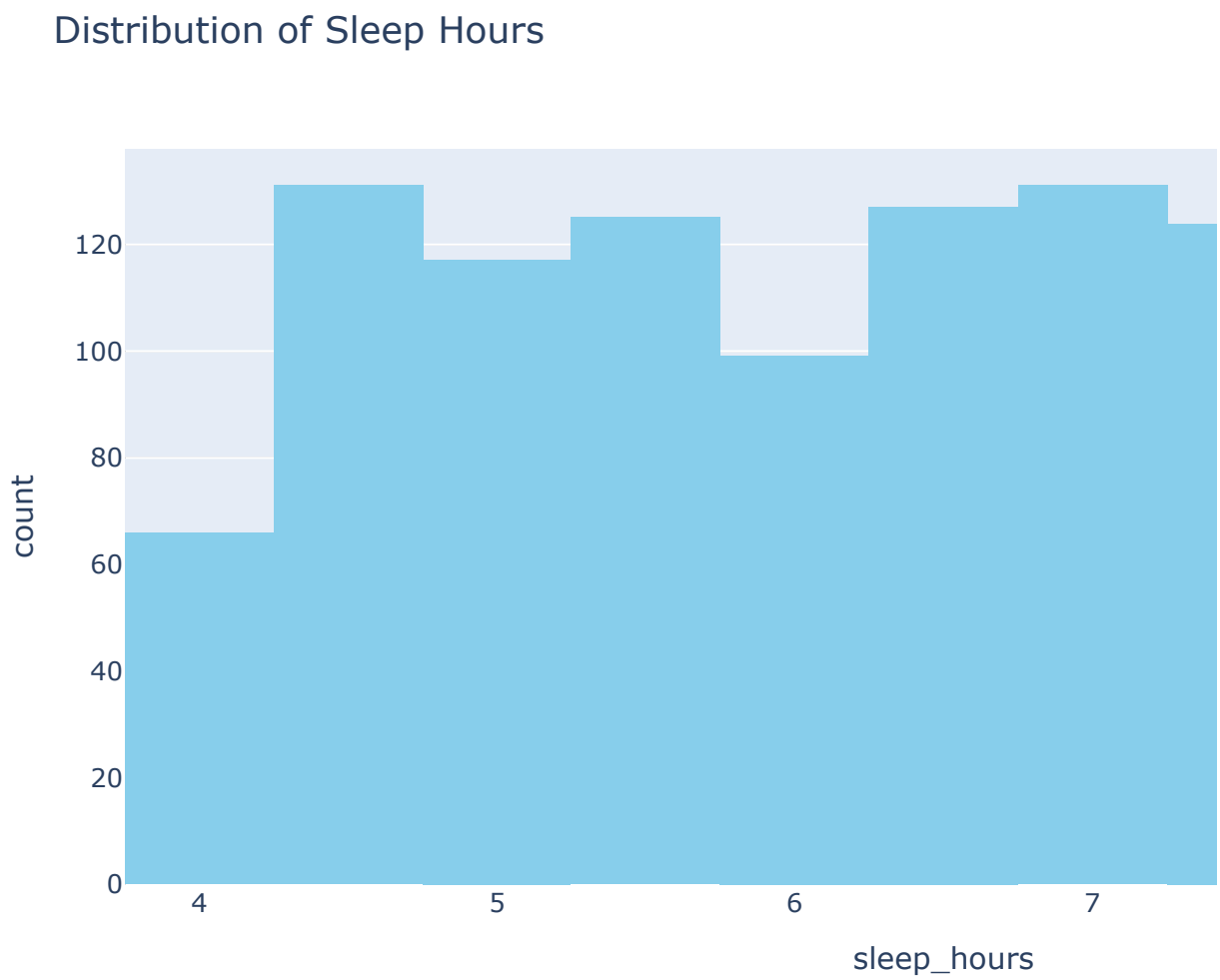
```
Index(['age', 'gender', 'daily_social_media_hours', 'platform_usage',
       'sleep_hours', 'screen_time_before_sleep', 'academic_performance',
       'physical_activity', 'social_interaction_level', 'stress_level',
       'anxiety_level', 'addiction_level', 'depression_label'],
      dtype='object')
```



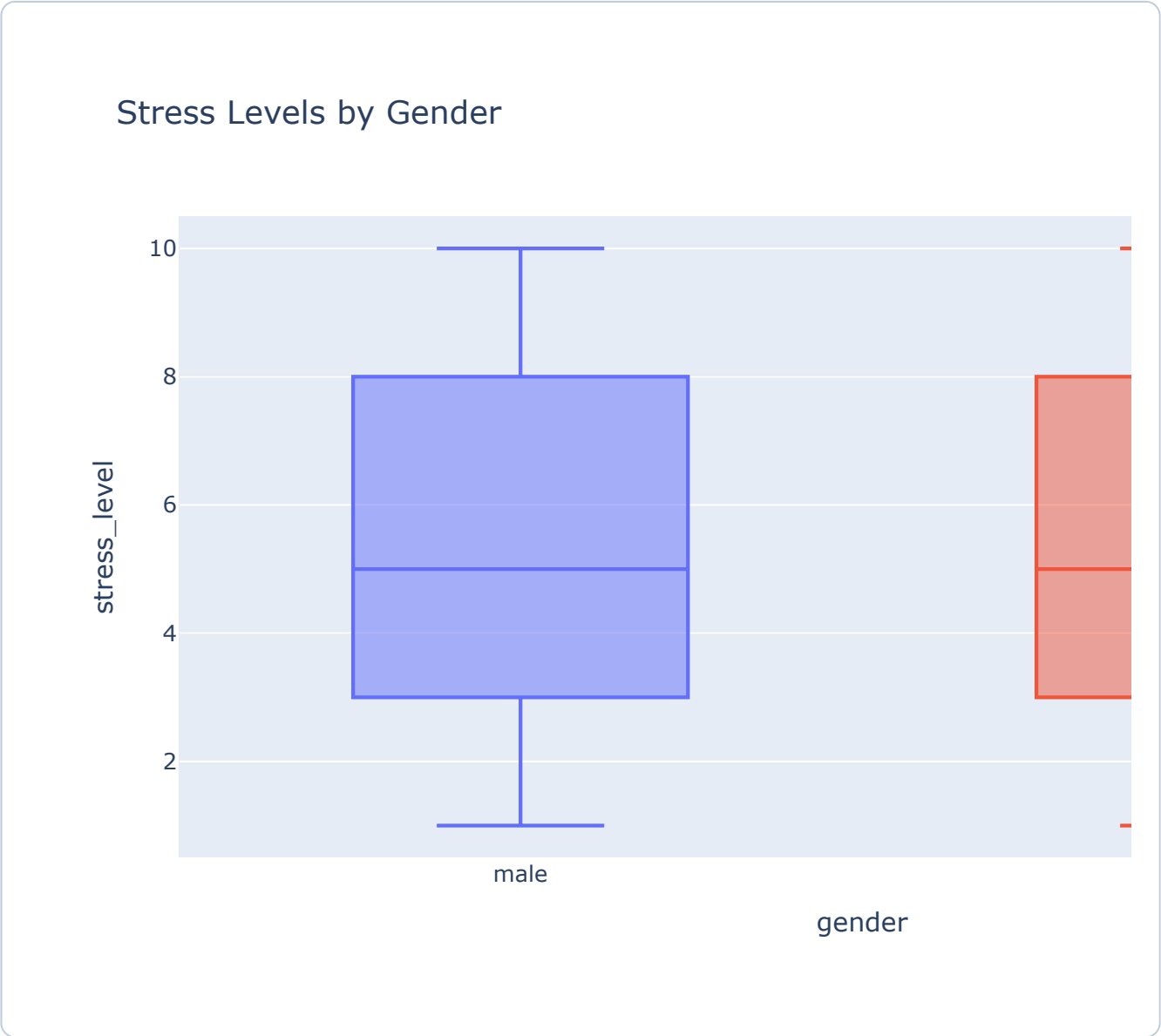
- - Higher stress, anxiety, and addiction levels tend to move together.
  - More sleep shows a mild positive link with academic performance and a mild negative link with stress.
  - Social media hours show small-to-moderate links with well-being and academics.
  Note: Correlations reflect associations, not causation.



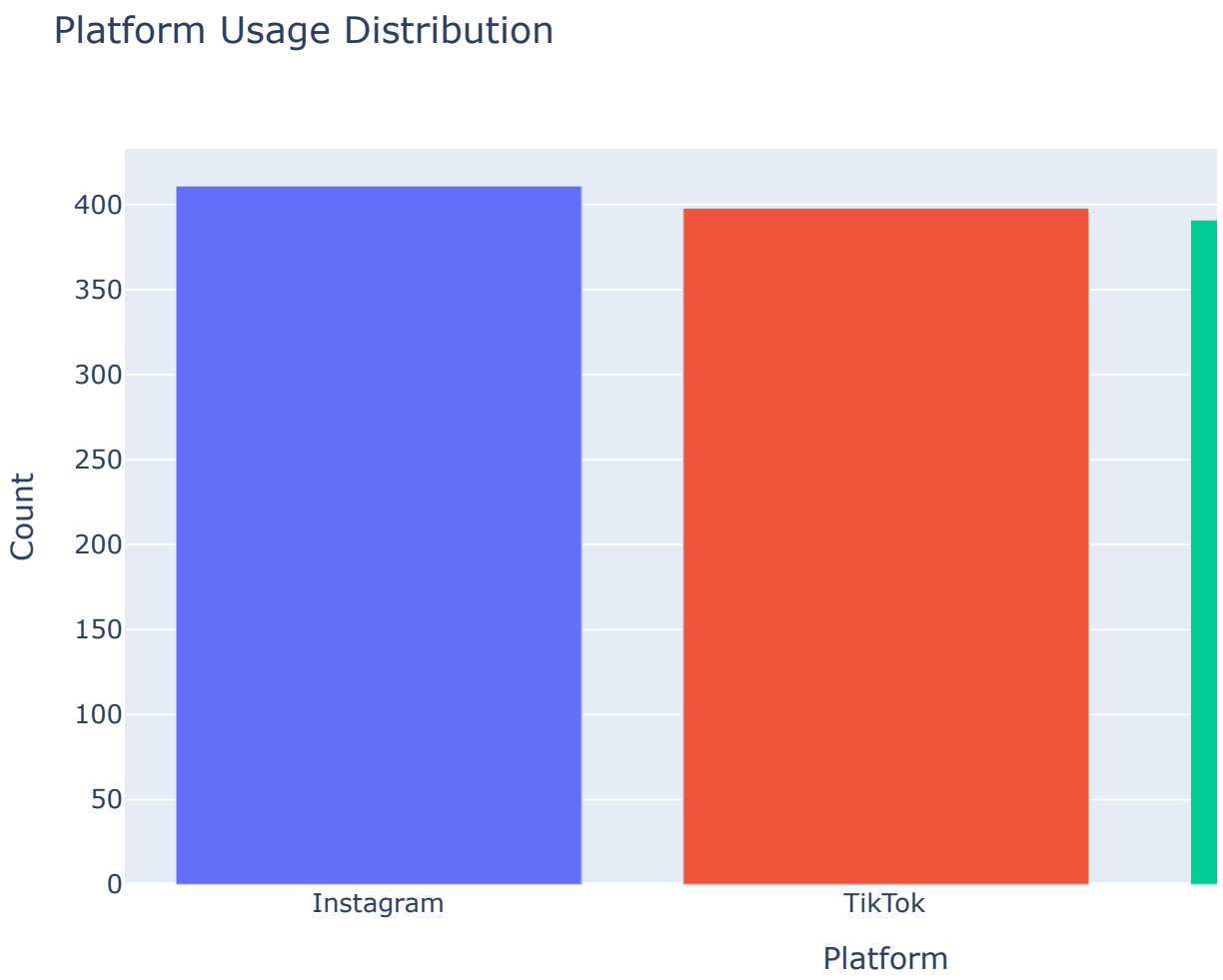
While higher social media hours show some skew among those labeled with depression, the overlap between groups is large.

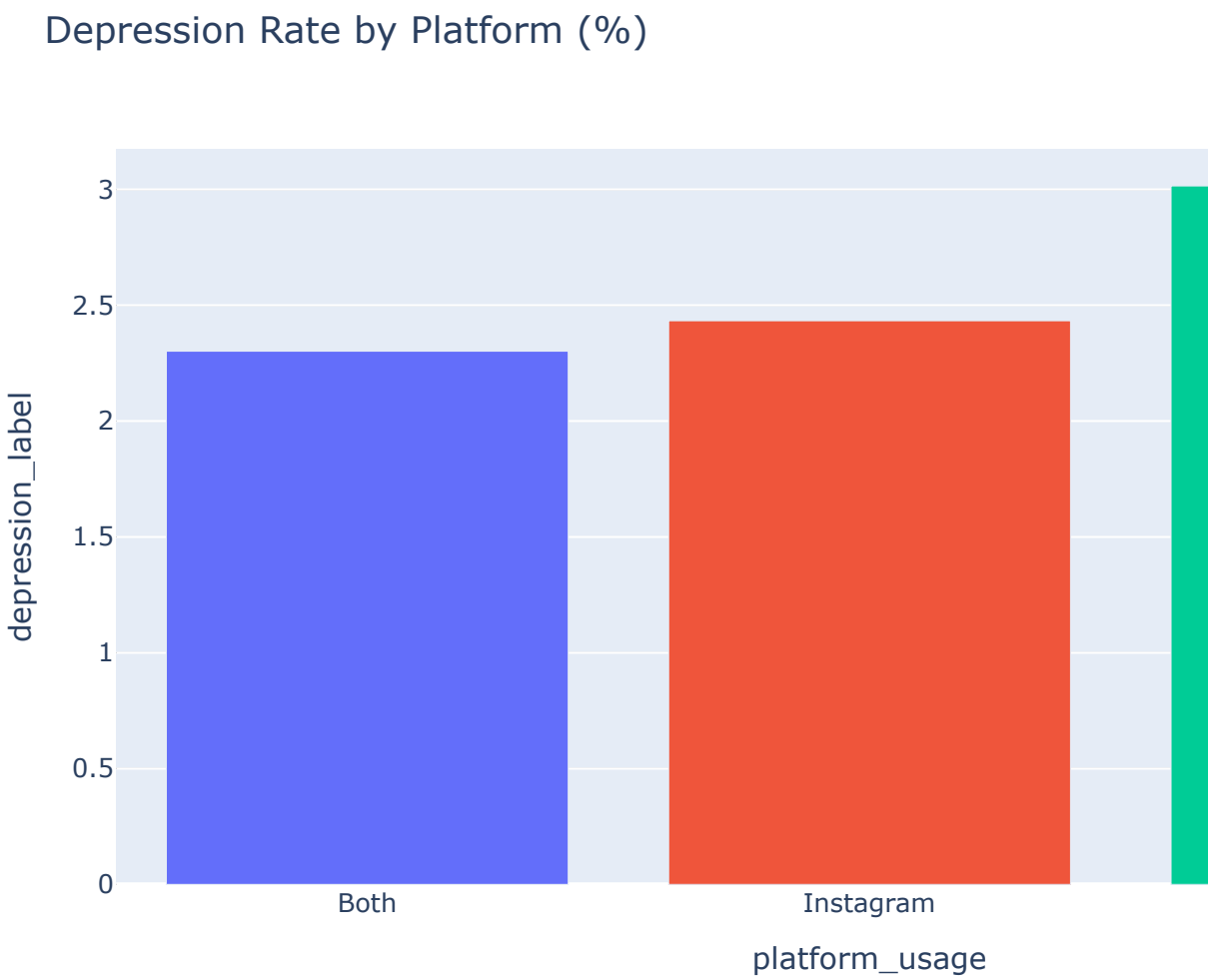


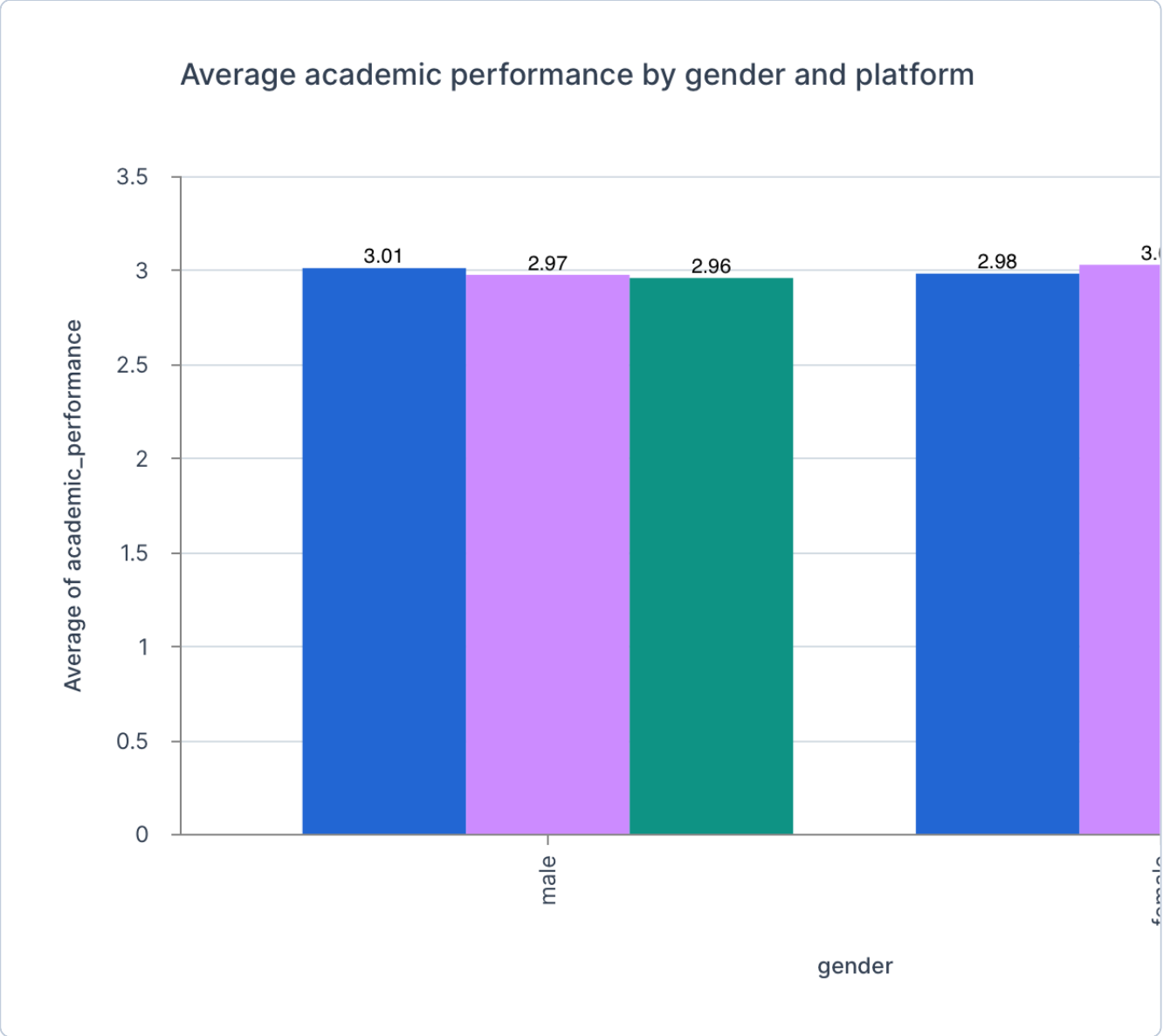
Sleep hours show a unimodal distribution centered around 6–7 hours, with most observations between roughly 5 and 8 hours.



Stress level differences by gender appear modest in this sample.


Platform Usage Distribution
Count
400
350
300
250
200
150
100
50
0
Instagram
TikTok
Platform


Depression Rate by Platform (%)
depression_label
3
2.5
2
1.5
1
0.5
0
Both
Instagram
platform_usage



Average academic performance varies little by gender, with only modest differences across platforms in this sample.

Compute basic data info: print shape, dtypes, missing values, and summary stats.

Dashboard: Teen Mental Health Summary

This lightweight dashboard summarizes key indicators from the Teen Mental Health dataset and provides a few quick visual explorations. Use it to scan overall patterns by gender, platform usage, and relationships between sleep and academic performance.

	Metric object	Value object	
0	Total rows	1200	
1	Avg daily social m...	4.54	
2	Avg sleep hours	6.45	
3	Avg academic pe...	2.99	
4	Avg stress level	5.45	
5	Avg anxiety level	5.64	
6	Avg addiction level	5.57	
7	Depression rate	2.58%	

8 rows, 2 cols 10 / page Page 1 of 1

Baseline: average social media use is about 4.5 hours/day, average sleep is about 6.5 hours/night, and the observed depression rate is roughly 2.6% in this sample.

Actionable summary & next steps

- - Associations between social media hours and sleep with academic performance appear small to moderate.
  - Stress, anxiety, and addiction levels tend to cluster together.
  - Gender differences are minor across most indicators in this sample.
  - Platform differences are subtle and may reflect sample composition.
  - Next steps: fit multivariate models (linear/logistic) with interactions (e.g., sleep × social media, platform × gender) to assess adjusted associations.

Below is a gender summary produced with DataFrame SQL for quick rollups.

	gender object	count int64	avg_daily_social_...	avg_sleep_hours f..	avg_acad
0	female	585	4.51	6.5	
1	male	615	4.56	6.4	

2 rows, 9 cols 10 / page Page 1 of 1